



△ Heart of the Crab Nebula

In the center of the Crab Nebula is a neutron star that is spinning 30 times a second and blasting out a blizzard of particles from its surface, as well as radiation beams from its poles. In this image taken by the Chandra X-ray Observatory, the ringlike structures around the pulsar (central blue-white dot) are shock waves produced where the wind of high-speed particles is plowing into the surrounding nebula.



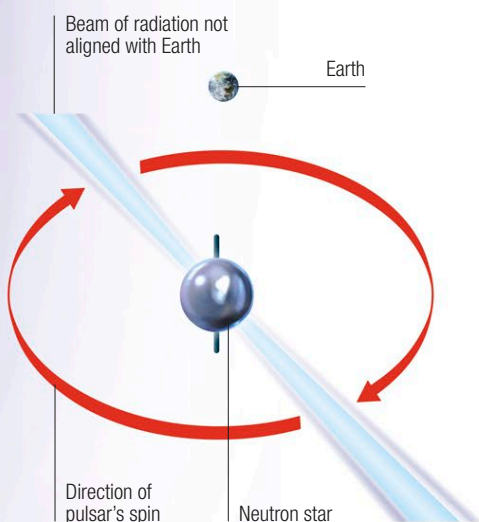
△ Pulsar 3C58

This image, taken with a camera that detects X-rays, shows the remains of an ancient supernova explosion. The bright central region, partially obscured by gas that emits X-rays (shown in blue), contains a pulsar. This is producing X-ray beams, which extend for trillions of miles to either side and have created loops and swirls (shown in blue and red) in other remnant material from the supernova.

Pulsars

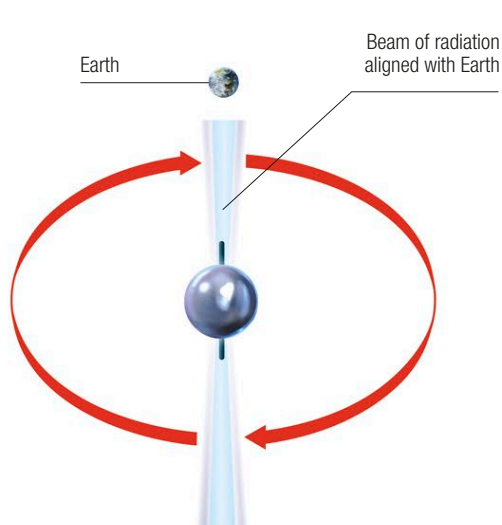
As they spin and sweep their radiation beams through space, neutron stars are like celestial lighthouses. If at least one of the radiation beams points toward Earth at some point in each rotation, then from Earth it will be detectable as a series of radiation pulses. Neutron stars that are detectable in this way are called “pulsars,” and the timing of their off/on signals have a precision comparable to that of an atomic clock. The first pulsar was discovered in 1967, but today more than 2,000 are known about in the Milky Way and nearby galaxies.

A neutron star’s gravity is so strong that it bends light emitted from its surface. So, if you could look at one you would see part of its far side as well as its near side



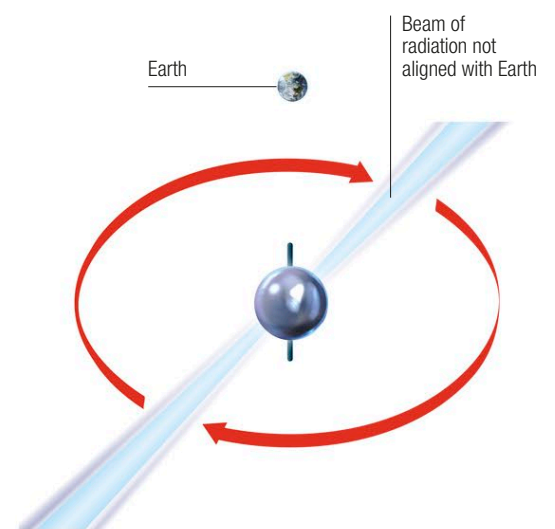
△ Pulsar off

As a pulsar rotates, its radiation beams continually sweep through space. At the instant shown here, neither beam points at Earth, so from the perspective of an observer on Earth, the pulsar is “off.”



△ Pulsar on

A moment later, one of the pulsar’s radiation beams is pointing at Earth. With the right equipment, this will be detectable on Earth as a brief signal or pulse of light, radio waves, X-rays, or other radiation.



△ Pulsar off

Very shortly afterward, the radiation beam is no longer aligned with Earth, so the pulse or signal switches “off” again. The off/on/off pulses occur at very regular intervals, characteristic of that pulsar.